

37. 设动点为 $P(x, y, z)$ 有: $(x-2)^2 + (y-2)^2 + (z-1)^2 = (x-1)^2 + (y-3)^2 + (z-4)^2$

$$\Rightarrow x^2 - 4x + 4 + y^2 - 4y + 4 + z^2 - 2z + 1 = x^2 - 2x + 1 + y^2 - 6y + 9 + z^2 - 8z + 16$$

$$\Rightarrow -4x - 4y - 2z + 9 = -2x - 6y - 8z + 26$$

$$\Rightarrow 2x - 2y - 6z + 17 = 0$$

38. 设 $M(x, y, z)$, $P(-R, 0, 0)$ $\vec{PM} = (x+R, y, z)$ $\vec{PM} \perp \vec{OM} \Rightarrow \vec{OM} \cdot \vec{PM} = 0$

$$\Rightarrow (x+R, y, z) \cdot (x, y, z) = x^2 + xR + y^2 + z^2 = 0$$

$$\Rightarrow M \text{ 轨迹为 } x^2 + xR + y^2 + z^2 = 0$$

39. (1) 抛物面. 

(2)

(4). $y = x^2 - z^2$ 为双曲抛物面 

41. (1) $x^2 + y^2 + z^2 = 1$

(3). $y^2 \pm (\sqrt{x^2 + z^2})^2 = 1 \Rightarrow y^2 - x^2 - z^2 = 1$

(4) $(\sqrt{z^2 + y^2})^2 = 5x \Rightarrow z^2 + y^2 = 5x$

42. (1) 为曲线 $\begin{cases} z^2 - \frac{y^2}{4} = 1 \\ x = 0 \end{cases}$ 形成, 为旋转双曲面.

13) 为曲线 $\begin{cases} \frac{1}{16}z^2 - y^2 = 9 \\ x=0 \end{cases}$ 开成. 为旋转双曲面

43.12) $y=-1$. $(x-1)^2 + (\frac{z}{2})^2 = 16$ 这是一个椭圆. 在 $y=-1$ 上, 中心为 $(1, -1, 0)$

44.11) 柱面: $x^2 + y^2 = 12$.

曲线: $\begin{cases} x^2 + y^2 = 12 \\ z=0 \end{cases}$

45.11)

46.11) $x = \cos t$, $\frac{y+1}{2} = \cos t$ $\frac{z}{3} = \sin t$

$\therefore x^2 + \frac{(y+1)^2}{4} + \frac{2}{9}z^2 = 2 \Rightarrow \frac{x^2}{2} + \frac{(y+1)^2}{8} + \frac{1}{9}z^2 = 1$

(2) $x-a=t$. $a^2 - y^2 = t$ $z^2 = 2a^2 - 2at \Rightarrow 2at = 2a^2 - z^2 \Rightarrow t = \frac{2a^2 - z^2}{2a}$

$\Rightarrow x^2 = a^2 + 2at + t^2$ $y^2 = a^2 - t^2$ $z^2 = 2a^2 - 2at$

$\therefore x^2 + y^2 + z^2 = 2a^2 + 2at + 2a^2 - 2at = 4a^2$

$\Rightarrow x^2 + y^2 + z^2 = 4a^2$

$$47.11) \quad x = a \sin t + x_0 \quad y = b \cos t + y_0 \quad z \in \mathbb{R}$$

$$\therefore \begin{cases} x = a \sin t + x_0 \\ y = b \cos t + y_0 \\ z = u \quad u \in \mathbb{R} \end{cases}$$

$$(2) \quad y = a \sec t \quad z = b \tan t \quad x = u \quad u \in \mathbb{R}$$

$$(3) \quad z = c \sec t \quad y = b \tan t \cos \theta \quad x = a \tan t \sin \theta$$

$$\therefore \begin{cases} x = a \tan t \sin \theta \\ y = b \tan t \cos \theta \\ z = c \sec t \end{cases}$$

$$(4) \quad x = u a \sin t + x_0 \quad y = u b \cos t + y_0 \quad z = u^2$$

$$(5) \quad x = a \sec t \sin \theta$$

$$y = b \sec t \cos \theta$$

$$z = c \tan \theta$$

$$1.4) \quad \cos \theta = \frac{x^2 + y^2 - z^2}{2xy} \Rightarrow \sin \theta = \sqrt{1 - \frac{(x^2 + y^2 - z^2)^2}{4x^2 y^2}}$$

$$\therefore S = \frac{1}{2} xy \sin \theta = \frac{1}{2} \sqrt{x^2 y^2 - \frac{(x^2 + y^2 - z^2)^2}{4}}$$

$$2.4) \quad 2x - x^2 - y^2 \neq 0 \Rightarrow x^2 + y^2 \neq 2x$$

$$(x^2 + y^2 - x)(2x - x^2 - y^2) \geq 0$$

$$\Rightarrow x \leq x^2 + y^2 < 2x$$

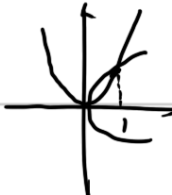
$$1.5) \quad y > x, \quad x \ln(y-x) > 0, \quad x \neq 0.$$

$$\therefore \begin{cases} y > x \\ x \ln(y-x) > 0 \end{cases}$$

$$1^\circ \text{ 若 } x > 0, \quad y-x > 1 \Rightarrow y > 1+x$$

$$2^\circ \text{ 若 } x < 0, \quad y-x < 1 \Rightarrow y < x+1$$

$$\therefore \begin{cases} \text{若 } x > 0, & y > 1+x \\ \text{若 } x < 0, & x < y < x+1 \end{cases}$$

$$3.17) \quad \begin{cases} y = x^y \\ y^2 = x \end{cases} \Rightarrow y = y^4 \Rightarrow y = 1$$


$$\therefore x \in [0, 1], \quad x^y \leq y \leq \sqrt{x}$$

$$\therefore \text{若 } x^2 \leq y \leq \sqrt{x}, \quad x \in [0, 1]$$

$$4.12) \quad f(x+y, x-y, xy) = (x+y)^{xy} + (xy)^{x+y}$$

$$13) \text{ 设 } f(x+y, \frac{y}{x}) = f(a, b)$$

$$x^2 y^y = (x+y)(x-y) = a(x-y)$$

$$\frac{y}{x} = b \Rightarrow y = bx$$

$$a = x+y = x+bx = x(1+b) \Rightarrow x = \frac{a}{1+b}, \quad y = \frac{ab}{1+b}$$

$$\therefore f(x+y, \frac{y}{x}) = x^2 y^y = \frac{a^y - a^y b^y}{(1+b)^y} = \frac{(1-b^y) a^y}{(1+b)^y} = \frac{(1-b) a^y}{1+b} = \frac{(1-\frac{y}{x})(x+y)^2}{1+\frac{y}{x}}$$

$$= \frac{(x-y)(x+y)^y}{x+y} = x^2 - y^y$$

6. (1) 令 $t = \sqrt{x+y+4}$. $x+y = t^2 - 4$

$$\therefore \lim_{t \rightarrow 2} \frac{2-t}{t^2-4} = \lim_{t \rightarrow 2} \frac{2-t}{(t-2)(t+2)} = \lim_{t \rightarrow 2} -\frac{1}{(t+2)} = -\frac{1}{4}$$

(2) 令 $xy = t$. $\lim_{(x,y) \rightarrow (0,0)} \frac{(2+x)\ln(1+xy)}{xy} = \lim_{t \rightarrow 0} \frac{2\ln(1+t)}{t} = 2 \lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 2$

(3) 令 $t = x^2 + y^2$. $(x,y) \rightarrow (0,0)$ 时 $t \rightarrow 0$

$$\therefore \lim_{t \rightarrow 0} \frac{\sin t}{t} = \sin 1$$

(4) 令 $t = \sqrt{x^2 + y^2}$. $(x,y) \rightarrow (0,0)$ 时 $t \rightarrow 0$

$$\lim_{t \rightarrow 0} t \sin \frac{1}{t} = 0$$

7. (2) 从 $y = x$ 趋近于 0 时, $\lim_{x \rightarrow 0} \frac{xy}{x+y} = \lim_{x \rightarrow 0} \frac{x^2}{2x} = 0$

从 $x+y = x^y$ 趋近 0. $\lim_{x \rightarrow 0} \frac{x(x^y - x)}{x^y} = \lim_{x \rightarrow 0} \frac{x^y - x}{x} = \lim_{x \rightarrow 0} (x - 1) = -1$

2个极限不同. 故无极限.

8. (1) $x^2 + y^2 = 0$ 处中断 $\Rightarrow$ 在 $(0,0)$ 处中断

(3) 在 $2x - y^2 = 0$ 处中断 $\Rightarrow y^2 = 2x$ 上中断

9. (1) $f = x+y - (x^2+y^2)^{\frac{1}{2}}$. $\frac{\partial f}{\partial x} = 1 + 0 - \frac{1}{2} \cdot \frac{1}{\sqrt{x^2+y^2}} \cdot 2x = 1 - \frac{x}{\sqrt{x^2+y^2}}$

$$\therefore f_x(3,4) = 1 - \frac{3}{5} = \frac{2}{5}$$

$$13) z = e^{y \ln(1+xy)}$$

$$\frac{\partial z}{\partial x} = e^{y \ln(1+xy)} \cdot \frac{y}{1+xy} \cdot y = \frac{y^2}{1+xy} e^{y \ln(1+xy)}$$

$$\left. \frac{\partial z}{\partial x} \right|_{(1,1)} = \frac{1}{2} e^{1 \cdot 1 \ln 2} = 1$$

$$\frac{\partial z}{\partial y} = e^{y \ln(1+xy)} \cdot [\ln(1+xy) + \frac{y}{1+xy} \cdot x]$$

$$\therefore \left. \frac{\partial z}{\partial y} \right|_{(1,1)} = 2 \cdot (\ln 2 + \frac{1}{2}) = 2 \ln 2 + 1$$

$$10.11) z = x \cdot (x^2 + y^2)^{-\frac{1}{2}}$$

$$\frac{\partial z}{\partial x} = (x^2 + y^2)^{-\frac{1}{2}} - \frac{1}{2} x (x^2 + y^2)^{-\frac{3}{2}} \cdot 2x = (x^2 + y^2)^{-\frac{1}{2}} - x^2 (x^2 + y^2)^{-\frac{3}{2}}$$

$$\frac{\partial z}{\partial y} = -\frac{1}{2} x (x^2 + y^2)^{-\frac{3}{2}} \cdot 2y = -xy (x^2 + y^2)^{-\frac{3}{2}}$$

$$c) z = \left(\frac{1}{3}\right)^{-\frac{y}{x}} = e^{\ln\left(\frac{1}{3}\right) \cdot \left(-\frac{y}{x}\right)} = e^{-\frac{y}{x} \ln 3} = e^{\frac{y}{x} \ln 3} = e^{\ln 3 \left(\frac{y}{x}\right)}$$

$$\frac{\partial z}{\partial x} = e^{\ln 3 \cdot \frac{y}{x}} \cdot \left(-\frac{y}{x^2}\right) \ln 3 = -\frac{\ln 3}{x^2} y \cdot 3^{\frac{y}{x}}$$

$$\frac{\partial z}{\partial y} = e^{\frac{y}{x} \ln 3} \cdot \left(\frac{\ln 3}{x}\right) = \frac{\ln 3}{x} \cdot 3^{\frac{y}{x}}$$

$$12.12) z = \arctan \frac{x+y}{1-xy}$$

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{1}{1 + \left(\frac{x+y}{1-xy}\right)^2} \cdot \frac{(1-xy) + y(x+y)}{(1-xy)^2} = \frac{1+y^2}{(1-xy)^2 + (x+y)^2} = \frac{1+y^2}{1-2xy+x^2y^2+x^2+2xy+y^2} \\ &= \frac{1+y^2}{1+x^2+y^2+x^2y^2} = \frac{1+y^2}{(1+y^2)(1+x^2)} = \frac{1}{1+x^2} \end{aligned}$$

$$\frac{\partial z}{\partial x^2} = -\frac{1}{(1+x^2)^2} \cdot 2x = \frac{-2x}{(1+x^2)^2}$$

$$\frac{\partial z}{\partial y^2} = \frac{-2y}{(1+y^2)^2}$$

$$\frac{\partial z}{\partial x \partial y} = \frac{\partial z}{\partial y \partial x} = 0$$

$$13.1) \frac{\partial z}{\partial x} = \ln xy + \frac{x}{xy} y = \ln xy + 1 = \ln x + \ln y + 1$$

$$\frac{\partial z}{\partial x} = \frac{1}{x} \quad \frac{\partial z}{\partial^2 x \partial y} = 0$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{1}{y}$$

$$\frac{\partial^2 z}{\partial x \partial y^2} = -\frac{1}{y^2}$$

$$13) \frac{\partial u}{\partial x} = 3x^2 - 3yz \quad \frac{\partial^2 u}{\partial x^2} = 6x$$

$$\frac{\partial u}{\partial y} = 3y^2 - 3xz \quad \frac{\partial^2 u}{\partial y^2} = 6y$$

$$\frac{\partial u}{\partial z} = 3z^2 - 3xy \quad \frac{\partial^2 u}{\partial z^2} = 6z$$

$$\therefore \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2 = 9x^4 - 18x^2yz + 9y^2z^2 + 9y^4 - 18y^2xz + 9x^2z^2 + 9z^4 - 18z^2xy + 9x^2y^2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 6(x+y+z)$$

$$14.1) \text{ ② } r = \sqrt{x^2 + y^2 + z^2} \quad \frac{\partial u}{\partial x} = \frac{-1}{r^2} \cdot \frac{x}{r} = -\frac{x}{r^3} \quad \frac{\partial^2 u}{\partial x^2} = \frac{3x^2 - r^2}{r^5}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{3y^2 - r^2}{r^5}$$

$$\frac{\partial^2 u}{\partial z^2} = \frac{3z^2 - r^2}{r^5} \quad \therefore \frac{3x^2 + 3y^2 + 3z^2 - 3r^2}{r^5} = 0$$

$\therefore$ 得 ②

$$13) \frac{\partial z}{\partial x} = \frac{1}{e^x + e^y} \cdot e^x = \frac{e^x}{e^x + e^y} \quad \frac{\partial z}{\partial y} = \frac{e^y}{e^x + e^y}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{e^x e^y}{(e^x + e^y)^2}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{e^x e^y}{(e^x + e^y)^2}$$

$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{e^x e^y}{(e^x + e^y)^2} \quad \Rightarrow \quad \frac{\partial^2 z}{\partial x^2} \cdot \frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial x \partial y} \quad \therefore \text{得 ②}$$

